AdBuster PRO — Architecture Overview (Public Preview)

This document provides a high-level overview of the internal architecture of AdBuster PRO. It does NOT include source code, algorithms, or implementation details. All processing is performed locally on the user's device.

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1. System Overview

AdBuster PRO is a Windows application that automatically stabilizes TV audio by detecting abnormal loudness patterns and sending infrared volume commands to a Broadlink device.

The system operates entirely offline and consists of three core components:

- Audio Engine (real-time audio analysis)
- CEPA Decision Engine (human-like volume logic)
- IR Control Engine (Broadlink integration)

No cloud services, accounts, or external servers are used.

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2. Audio Engine (High-Level Description)

The audio engine continuously monitors the incoming audio signal and extracts simplified metrics such as:

- smoothed loudness level
- short-term variations
- long-term stability indicators

These metrics are used by the decision engine to determine whether the current audio resembles normal program content or a commercial break.

Implementation details and formulas are proprietary.

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3. CEPA Decision Engine (Conceptual Overview)

CEPA (Contextual Event Pattern Analysis) is a rule-based decision layer that mimics human perception of loudness changes.

Its responsibilities include:

- interpreting audio patterns
- distinguishing natural fluctuations from disruptive spikes
- determining when a volume adjustment is necessary
- preventing over-correction
- restoring volume smoothly after loud segments

Exact logic, thresholds, and algorithms are proprietary.

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4. Machine Learning Layer (PRO Version)

The PRO version includes an enhanced ML model that improves detection accuracy by analyzing long-term audio behavior.

The ML layer:

- processes simplified audio metrics
- identifies patterns typical for commercial breaks
- assists CEPA in decision-making

The model runs locally and does not upload or transmit any data.

Model architecture and training methods are proprietary.

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5. IR Control Engine (Broadlink Integration)

AdBuster PRO communicates with Broadlink IR devices to send:

- volume down
- volume up
- mute

The IR engine includes:

- device discovery
- IR code learning
- command throttling
- connection watchdog

Low-level IR handling and server implementation are proprietary.

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6. User Interface (Overview)

The application includes a Windows GUI that provides:

- real-time audio visualization
- threshold controls
- calibration tools
- mode selection (AUTO / ML / MUSIC / AD)
- Broadlink device status

The interface is built using standard Windows UI frameworks.

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7. Privacy & Security

AdBuster PRO is fully offline:

- no cloud
- no telemetry
- no audio recording
- no external communication

All processing happens locally on the user’s PC.

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8. Licensing & PRO Features

The PRO version unlocks:

- advanced ML model
- extended history analysis
- additional CEPA features
- enhanced stability logic

The full source code is proprietary and not publicly available.

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